21ECC301P - MICROPROCESSOR,
MICROCONTROLLER AND INTERFACING TECHNIQUES

SEMESTER V
(2025-26 ODD)

S.P.I.D.E.X. – Spider Platform for Integrated Dynamic Engineering X-bot

A PROJECT REPORT

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NOVEMBER 2025

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BONAFIDE CERTIFICATE

Certified that this project report **S.P.I.D.E.X. – Spider Platform for Integrated Dynamic Engineering X-bot** in the Bonafide work of **Parnapalli Anish [RA2311004010117], Tavishi Bhardwaj [RA2311004010119], T.S.Kubera Vishnu [RA2311004010121]** who carried out the project work under my supervision, as a part of the course **"Microprocessor**, **Microcontroller and Interfacing Techniques"** during the academic year 2025-26 (ODD) in the Department of ECE, SRM Institute of Science & Technology, Kattankulathur.

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DECLARATION

We hereby declare that the Project entitled “S.P.I.D.E.X. – Spider Platform for Integrated Dynamic Engineering X-bot” to be submitted for the course “21ECC301P – Microprocessor, Microcontroller and Interfacing Techniques”, is our original work as a team

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ACKNOWLEDGEMENT

We would like to express our deepest gratitude to the entire management of SRM Institute of Science and Technology for providing us with the necessary facilities for the completion of this project.

We wish to express our deep sense of gratitude and sincere thanks to our Professor and Head of the Department DR. M. SANGEETHA, for her encouragement, timely help, and advice offered to us. We wish to express our sincere thanks to our professor in-charge DR. P. ARUNA PRIYA, for her constant motivation.

We would like to express our deepest gratitude to our guide, Dr. P. ESWARAN for her valuable guidance, consistent encouragement, personal caring, timely help and providing me with an excellent atmosphere for doing research. All through the work, in spite of her busy schedule, she has extended cheerful and cordial support to us for completing this research work.

We would like to express our sincere thanks to our faculty coordinators Dr. P. ESWARAN and Dr. S.GIRI PRASAD for their time and suggestions for the implementation of this project.

We also extend our gratitude and thanks to all the teaching and non-teaching staff of the Electronics and Communication Engineering Department and to my parents and friends, who extended their kind cooperation using valuable suggestions and timely help during this project work.

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ABSTRACT

This project presents the design, construction, and control of SPIDEX, an untethered quadruped spider robot developed for adaptive locomotion across uneven terrain. The robot employs a 12-servo actuation system (three per leg) to replicate the coordinated joint movement of arachnids. At its core, an Arduino Nano microcontroller mounted on an expansion board executes real-time inverse kinematics, translating directional commands into precise servo angles for stable walking, turning, and crouching gaits. Wireless control is achieved through an HC-05 Bluetooth interface paired with a mobile app, enabling seamless user operation and feedback.

A critical engineering challenge addressed is the power architecture. The system integrates a dual-rail configuration using two Li-ion batteries and an XL4016 buck converter, providing a stable 6 V supply to manage simultaneous high-current servo demands while preventing controller brownouts. Experimental validation confirmed reliable motion, smooth gait transitions, and consistent wireless control. Aligning with SDG 9 (Industry, Innovation and Infrastructure) and SDG 11 (Sustainable Cities and Communities), SPIDEX demonstrates the potential of affordable, bio-inspired robotics for exploration and disaster-response applications.

SDG 9 - Industry, Innovation and Infrastructure

SDG 11 - Sustainable Cities and Communities

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CHAPTER 1
INTRODUCTION

1.1. Introduction

The field of mobile robotics has witnessed rapid advancements in recent years, driven by the growing demand for autonomous systems capable of performing complex tasks in challenging and unstructured environments. Traditional wheeled robots, though efficient on smooth surfaces, encounter severe limitations when navigating rough, debris-filled, or uneven terrains such as disaster zones, forests, or rocky landscapes. These constraints have inspired the development of legged robots, which draw inspiration from biological organisms to achieve higher mobility and adaptability. Among them, quadruped or spider-like robots have gained particular interest for their ability to balance, climb, and maneuver across surfaces where wheeled motion fails.

This project, titled S.P.I.D.E.X. – Spider Platform for Integrated Dynamic Engineering X-bot, focuses on designing and implementing a compact, low-cost quadruped robot capable of wireless, app-controlled locomotion. The robot is designed around a 12-servo configuration, providing three degrees of freedom per leg, which enables life-like motion similar to natural arachnids. The system integrates an Arduino Nano microcontroller as the primary processing unit, supported by an HC-05 Bluetooth module for communication and a dual-rail power system using Li-ion batteries and a buck converter to ensure reliable operation. Through the integration of mechanical design, electronics, and embedded control algorithms, the SPIDEX robot demonstrates how affordable hardware and open-source tools can be combined to build intelligent robotic platforms.

The importance of this project lies in its multidisciplinary nature and its relevance to modern engineering challenges. It not only strengthens understanding of mechatronic integration, embedded systems, and inverse kinematics but also promotes innovation aligned with Sustainable Development Goal (SDG) 9 – Industry, Innovation and Infrastructure, and SDG 11 – Sustainable Cities and Communities. By advancing low-cost, mobile robotic systems, SPIDEX contributes to technological education and paves the way for future applications in search and rescue, environmental monitoring, and exploration.

This report provides a detailed account of the design, development, and testing of the SPIDEX robot. It begins with a literature review highlighting existing quadruped research, followed by the system's hardware and software architecture, methodology, and implementation process.

1.2. Problem statement

The main problem addressed by the S.P.I.D.E.X. (Spider Platform for Integrated Dynamic Engineering X-bot) project is the limitation of traditional wheeled robots in navigating uneven and unstructured terrains. Wheeled robots, while efficient on smooth surfaces, struggle to maintain stability and mobility when confronted with obstacles such as debris, stairs, rocky ground, or natural terrain irregularities—conditions commonly encountered in disaster relief zones, exploration sites, or remote environments. This inherent limitation restricts their usability in real-world scenarios where flexibility and adaptive motion are crucial. To overcome this challenge, the S.P.I.D.E.X. project aims to design and develop a bio-inspired quadruped robotic platform that mimics the multi-legged locomotion of spiders. The project focuses on implementing a twelve-servo actuation system controlled via inverse kinematics algorithms, enabling stable walking gaits in multiple directions. Additionally, it integrates a Bluetooth-based wireless control system for untethered operation and a dual-rail power management system to ensure consistent performance without voltage drops or controller resets. Overall, the problem being addressed is how to create a cost-effective, stable, and educational quadruped robot capable of adaptive, multi-directional movement in real-time, thereby bridging the gap between complex research-grade robots and accessible educational robotics platforms.

1.3. Objective

- To design and construct a functional 12-servo quadruped robot (SPIDEX) capable of app-controlled wireless locomotion.
- To implement robust inverse kinematics and Bluetooth communication for real-time control and movement.
- To engineer a stable, dual-rail power system ensuring reliable operation for both logic and high-current servo actuation.

CHAPTER 2
LITERATURE SURVEY

2.1. An Overview of Model Predictive Optimization and Control of Quadruped Whole-Body Locomotion

C. Cun, Q. Yang, Z. Li, M. C. Zhou, and J. Pang (2025) presented their research titled "Model Predictive Optimization and Control of Quadruped Whole-Body Locomotion" in the IEEE/CAA Journal of Automatica Sinica. The authors introduced a two-layer Model Predictive Control (MPC) framework that combines Ground Reaction Force (GRF) optimization with motion and force control for maintaining stability during dynamic locomotion. Their work demonstrated that the use of predictive and machine learning-based solvers could significantly enhance balance, adaptability, and energy efficiency in quadruped robots. The MPC framework enabled smooth gait transitions such as crawling and trotting while maintaining stability on uneven terrains.

This study is highly relevant to the SPIDEX project, as it provides a theoretical foundation for future enhancement of its open-loop gait system into a closed-loop predictive model. While SPIDEX currently uses a simpler inverse kinematics and sequence-based motion approach, the implementation of MPC in later versions could allow adaptive gait control, better balance correction, and real-time terrain response — essential features for next-generation quadrupeds.

2.2. Quadruped Robots: Bridging Mechanical Design, Control, and Applications

Q. Li et al. (2025), in their comprehensive review "Quadruped Robots: Bridging Mechanical Design, Control, and Applications" published in MDPI (Robots), analyzed the key aspects of modern quadruped robots, including mechanical design, actuation methods, control algorithms, and artificial intelligence integration. The review emphasized that the success of quadruped locomotion lies in achieving a seamless coordination between mechanical structure, control electronics, and power delivery. The paper also discussed various approaches to improve stability, energy consumption, and gait adaptability through intelligent motion control.

The findings of this research strongly influenced the design approach adopted in SPIDEX. The SPIDEX system focuses on achieving a balance between performance and cost by using compact, lightweight mechanical structures, efficient servo actuation, and modular electronic integration. The study also validated the importance of inverse kinematics as a reliable method for controlling multi-jointed robotic legs, which is central to SPIDEX's motion control strategy.

2.3. OpenMutt: A Reconfigurable Quadruped Robot for Research and Education

M. R. Garcia et al. (2024), in their article "OpenMutt: A Reconfigurable Quadruped Robot for Research and Education" published in the International Journal of Mechanical Engineering Education, introduced OpenMutt, an open-source quadruped robot designed for academic and experimental use. The system was developed with modular hardware components, making it easy to reconfigure, test, and modify. The researchers highlighted the need for low-cost educational platforms that allow students to understand robotic locomotion principles without requiring expensive or complex hardware.

This study is particularly relevant to SPIDEX, as it validates the concept of creating an accessible and modular quadruped platform for learning and experimentation. Similar to OpenMutt, SPIDEX utilizes affordable components such as Arduino Nano, HC-05 Bluetooth module, and SG90 servos, which make it suitable for academic projects. Furthermore, OpenMutt's modular design inspired the mechanical flexibility and scalability incorporated into SPIDEX, allowing future upgrades like advanced sensor integration or improved servo mechanisms.

2.4. Sample-Based Stochastic Control Strategies for Quadrupedal Robots

G. Turrisi et al. (2024), in their preprint "Sample-Based Stochastic Control Strategies for Quadrupedal Robots", explored an alternative approach to deterministic control by introducing sample-based stochastic control methods. Their work emphasized the importance of handling uncertainty in actuator performance and environmental conditions. By simulating noise and random variations, the stochastic control strategy improved the robot's adaptability and robustness, allowing quadrupeds to maintain stability even on uneven terrains or under external disturbances.

This research is relevant to SPIDEX because it provides insights into how adaptive and noise-tolerant control methods can enhance real-world performance. While SPIDEX currently operates on pre-programmed inverse kinematic sequences, integrating stochastic or sensor-assisted feedback systems in future iterations could help the robot adjust its movement dynamically to unpredictable terrain or load changes. Such advancements could transform SPIDEX from a static gait-based platform into a more intelligent and environment-aware quadruped system.

CHAPTER 3
SYSTEM DESCRIPTION AND METHODOLOGY

3.1. Block Diagram

The block diagram illustrates the overall hardware architecture of the SPIDEX quadruped robot, showing the interconnection between the power supply, control, communication, and actuation units. Each block represents a major subsystem contributing to the robot's functionality and stability.

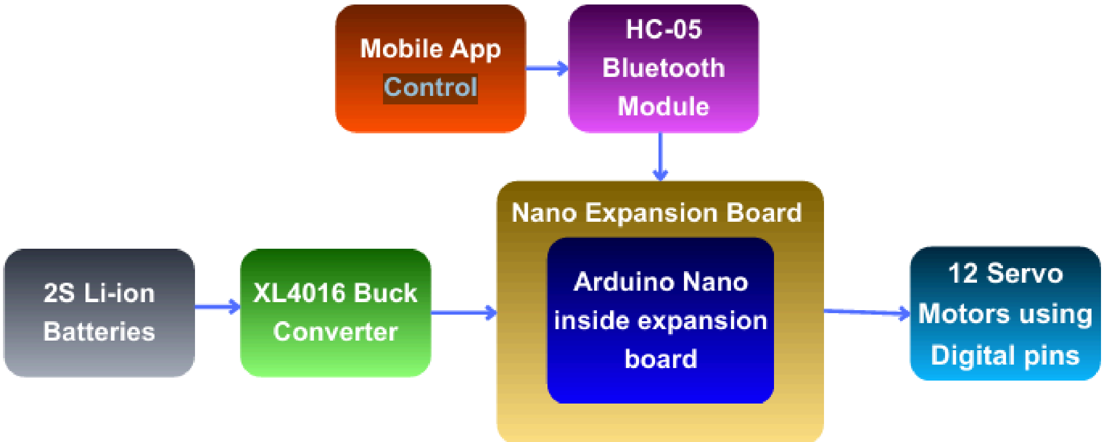

Figure 3.1. Block Diagram of S.P.I.D.E.X

Power Supply Unit (2S Li-ion Batteries)

The robot is powered by a 2S Li-ion battery pack (nominal voltage 7.4 V). These batteries provide the necessary energy to drive both the microcontroller and multiple servos. The use of Li-ion cells ensures high energy density, light weight, and rechargeable operation, which are ideal for mobile robotic platforms.

Voltage Regulation (XL4016 Buck Converter)

The XL4016 Buck Converter steps down the 7.4 V battery voltage to a stable 6 V output suitable for the SG90 servo motors and control circuits. This converter also provides current regulation to prevent voltage drops during high-load conditions when multiple servos operate simultaneously. It ensures safe and reliable operation without overheating or brownouts.

Control Unit (Arduino Nano on Nano Expansion Board)

At the core of SPIDEX lies the Arduino Nano, which serves as the main controller. It is mounted on a Nano Expansion Board to simplify wiring, provide additional power rails, and organize I/O pin connections.

- The Arduino Nano executes the Inverse Kinematics (IK) algorithms that calculate precise joint angles for each of the 12 servos.
- It generates PWM (Pulse Width Modulation) signals through its digital pins to control the angular position of every servo motor.
- The Nano also manages Bluetooth communication, command processing, and gait sequence execution.

Communication Unit (HC-05 Bluetooth Module)

The HC-05 Bluetooth Module provides wireless communication between the robot and a mobile application. It operates using a serial interface (TX/RX pins) connected to the Arduino Nano. Through this link, the user can send commands (e.g., Forward, Backward, Left, Right, Stop) from the app, which are decoded by the Nano to trigger corresponding gait movements.

This ensures real-time, untethered control of the robot.

Mobile Application Control

The Mobile App serves as the user interface for robot operation. Built using a Bluetooth terminal or a custom RC control app, it transmits directional commands to the HC-05 module. This allows the user to manually control the robot's movement from a smartphone with simple button presses.

Actuation Unit (12 Servo Motors)

The 12 SG90 servo motors form the robot's legs — three per leg (coxa, femur, and tibia joints). Each servo receives a PWM control signal from the Arduino Nano, defining its angular position. The coordinated control of all 12 servos enables the robot to perform walking, turning, and crouching movements.

Overall Operation Summary

- The Li-ion battery supplies power through the buck converter, providing stable voltage to the controller and servos.
- The Arduino Nano runs motion algorithms and generates control signals for the servos.
- The HC-05 module receives commands from the mobile app and relays them to the Nano via Bluetooth.
- The Nano translates these commands into servo movements, allowing smooth and stable locomotion of the quadruped robot.

3.2. Pin Diagram

Detailed Pin Explanation – Arduino Nano Connections:

D0 (RX) & D1 (TX):

- These pins are used for serial communication with the HC-05 Bluetooth module.
- D0 (RX) receives data from the Bluetooth module's TX pin.
- D1 (TX) transmits data to the Bluetooth module's RX pin.
- This setup enables two-way wireless communication between the mobile app and the robot.

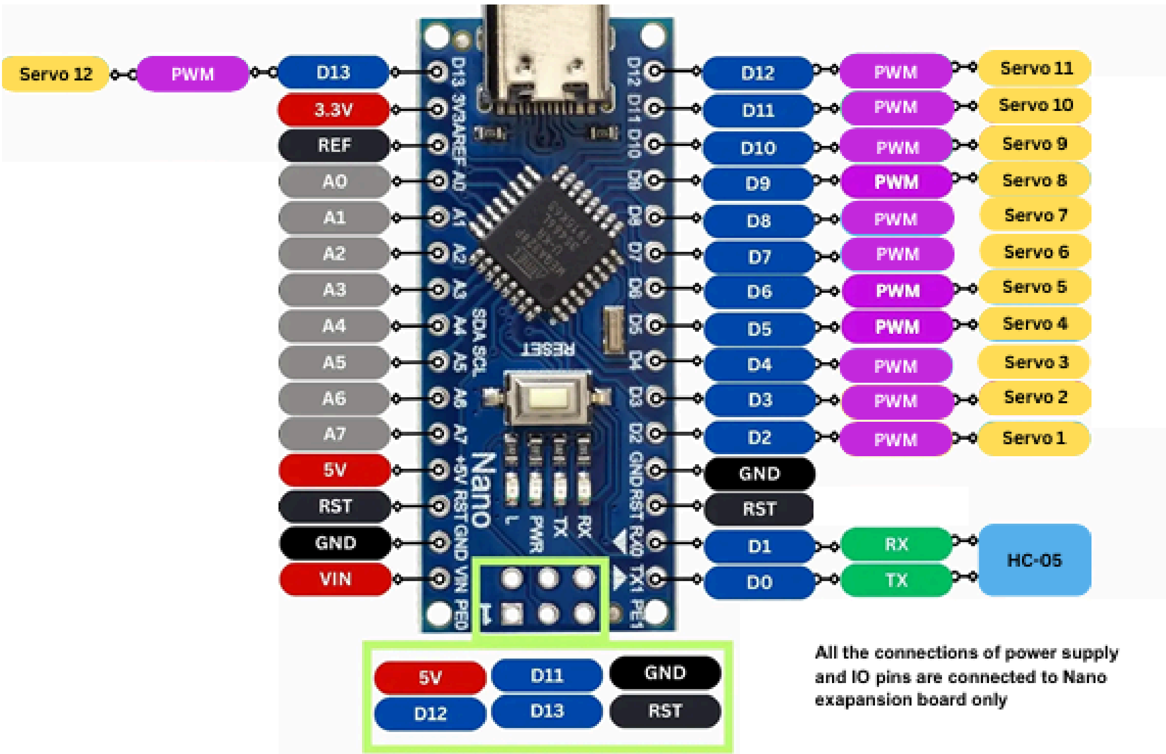

Figure 3.2. Arduino Nano Pin Diagram

D2 – D13 (PWM Pins):

These 12 digital pins are configured to control the 12 servo motors. Each servo is connected to one pin:

Servo 1 → D2	Servo 7 → D8
Servo 2 → D3	Servo 8 → D9
Servo 3 → D4	Servo 9 → D10
Servo 4 → D5	Servo 10 → D11
Servo 5 → D6	Servo 11 → D12
Servo 6 → D7	Servo 12 → D13

5V Pin:

Supplies regulated 5V power from the buck converter to the Arduino Nano and the connected sensors or modules. It powers the logic circuit and provides stable voltage for control operations.

VIN Pin:

Connected to the output of the Li-ion battery pack (7.4V) through the Nano expansion board. It is internally regulated to 5V by the Arduino's onboard voltage regulator, ensuring safe operation of logic circuits.

GND (Ground) Pins:

All ground connections from the servo motors, HC-05 module, buck converter, and battery are tied together here. A common ground reference is essential for consistent voltage levels and smooth servo operation.

3.3V Pin:

Provides a 3.3V output (if required) for any low-voltage modules or future sensors, though it is not used in the current SPIDEX configuration.

RST Pin:

Connected to the reset pin on the Nano expansion board. It allows manual or programmatic reset of the Arduino when required.

A0 – A7 (Analog Pins):

These pins are available for future use such as connecting sensors (e.g., ultrasonic, IMU, or force sensors) for advanced control. They are not used in the current design but reserved for upgrades.

Summary of Functionality

D0–D1: Bluetooth communication

D2–D13: Servo control (PWM)

5V, VIN, GND: Power distribution

RST: System reset

A0–A7: Reserved for future sensor integration

3.3.Flow Diagram

The Flow diagram of the SPIDEX robot illustrates the complete integration of the power, control, communication, and actuation systems required for smooth quadruped locomotion. The power source consists of two Li-ion batteries connected in series, providing a nominal 7.4V supply. This voltage is fed into an XL4016 buck converter, which steps down the voltage to a stable 6V rail dedicated to powering all 12 servo motors. The buck converter is rated for 8A output, ensuring sufficient current delivery during simultaneous servo operations. The Arduino Nano, mounted on an expansion board, receives its power from the same battery pack through the VIN pin, which is internally regulated to 5V for the controller and logic circuits.

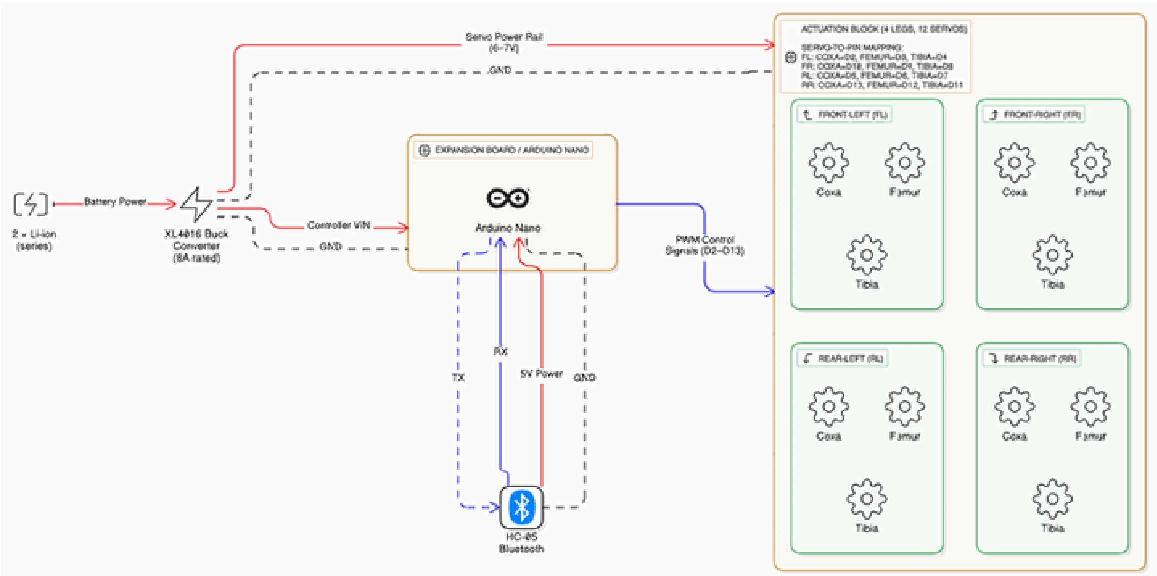

Figure 3.3. Flow Diagram of S.P.I.D.E.X

The HC-05 Bluetooth module interfaces with the Arduino via TX (D1) and RX (D0) pins, enabling wireless serial communication between the robot and a mobile control app. Commands such as forward, backward, left, and right are transmitted via Bluetooth and processed by the Arduino. The controller then sends PWM control signals (D2–D13) to the actuation block, which includes four legs—Front Left (FL), Front Right (FR), Rear Left (RL), and Rear Right (RR)—each having three servos corresponding to the coxa, femur, and tibia joints. These servos perform synchronized motion through inverse kinematic calculations to achieve walking, turning, and balancing movements. The GND lines of all modules are interconnected to maintain a common reference point, ensuring system stability. Overall, this architecture enables efficient power management, precise motion control, and seamless Bluetooth-based wireless operation, making SPIDEX a robust and mobile quadruped platform.

3.4. Kinematics And Calibrations

Kinematics of SPIDEX Leg Mechanism:

The kinematics of the SPIDEX robot defines the mathematical relationship between the servo joint angles and the position of the leg's end effector (foot) in 3D space. Each leg consists of three servos — corresponding to the coxa, femur, and tibia joints — providing three degrees of freedom (DOF), which allow the leg to move in the x, y, and z directions.

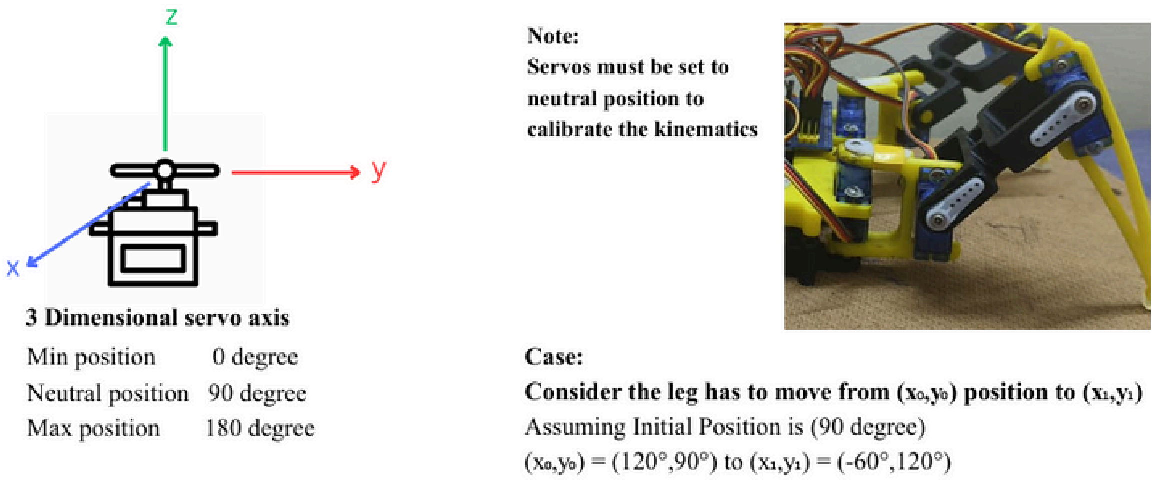

Figure 3.4. Kinematics And Calibrations

Servo Axis Orientation:

As shown in the diagram, the three servo axes correspond to a 3D coordinate system:

X-axis – Lateral movement (sideways swing of the leg).

Y-axis – Forward and backward leg extension.

Z-axis – Vertical lift or lowering of the leg.

Each servo has a rotational range from 0° (minimum) to 180° (maximum), with 90° defined as the neutral position. This neutral angle represents the centered or resting position of each joint, where no displacement is applied.

Calibration:

Before performing any motion or gait testing, each servo must be set to its neutral position (90°). This ensures that the physical position of the leg matches the reference position in the kinematic model.

If this step is skipped, the leg will not align correctly with the calculated motion paths, resulting in unstable or incorrect gait movements.

Example Case

Consider a single leg moving from an initial position (x_0, y_0) to a target position (x_1, y_1) in the 2D plane (for simplicity, ignoring z-motion):

Initial joint angles: (x_0, y_0) = (120°, 90°)

Final joint angles: (x_1, y_1) = (–60°, 120°)

This means the leg rotates horizontally by 180° (from +120° to –60°) while simultaneously extending vertically from 90° to 120°, simulating a forward stride. The inverse kinematics equations calculate the required angles for each servo to move the foot from its starting point to the desired end point smoothly.

Inverse Kinematics Principle

Inverse Kinematics (IK) computes the servo angles (θ_1, θ_2, θ_3) needed to reach a specific foot position (x, y, z).

For each leg:

x=L1cos(θ1)+L2cos(θ1+θ2)x = L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2)x=L1 cos(θ1)+L2cos(θ1+θ2) y=L1sin(θ1)+L2sin(θ1+θ2)y = L_1 \sin(\theta_1) + L_2 \sin(\theta_1 + \theta_2)y=L1sin(θ1)+L2sin(θ1+θ2)

where:

L_1, L_2 = link lengths (femur and tibia segments)

θ_1, θ_2 = servo angles at coxa and femur joints

By rearranging these equations, the microcontroller computes the servo angle commands for a given foot coordinate, allowing real-time control of the robot's gait.

Application in SPIDEX

In SPIDEX, these kinematic calculations are embedded within the Arduino Nano program. The controller sends appropriate PWM signals to each servo, moving them to calculated angles to achieve forward, backward, turning, or gestural movements.

Each leg's IK function operates independently but synchronizes through the main gait engine, producing stable and smooth walking patterns.

3.5. Tools/Software Used

The development of the SPIDEX robot required a combination of CAD tools for mechanical design, an Integrated Development Environment (IDE) for programming the microcontroller, and specific software components for wireless control.

Table 3.1. Tools/Software Used

Category	**Software/Tool**	**Purpose in Project**
Microcontroller IDE	Arduino IDE	Used for writing, compiling, and uploading the C++ firmware (containing the IK and gait engine logic) onto the Arduino Nano.
Programming Language	C/C++	The primary language used for implementing the complex Inverse Kinematics equations and real-time control logic on the constrained hardware environment of the Arduino.
Robotics Library	Servo.h Library	Standard Arduino library used to generate the necessary Pulse Width Modulation (PWM) signals on digital pins (D2-D13) to control the 12 servo motors.
Communication	Mobile RC App (Generic)	A customizable, commercially available or open-source Bluetooth RC application used on an Android or iOS device to provide the user interface for sending simple command characters ('F', 'B', 'L', 'R', etc.) to the HC-05 module.
Mechanical Design	Fusion 360 / SolidWorks (Assumed)	Used for the design and dimensioning of the custom 3D-printed chassis parts and leg segments, ensuring proper integration of the SG90 servos.

CHAPTER 4
RESULTS AND DISCUSSION

4.1. Output Screenshots

The S.P.I.D.E.X. robot was successfully constructed, and its functionality was validated through a series of mobility and power system tests.

The robot's movements, including forward, reverse, turning, and side walking, were observed to be smooth and well-coordinated under Bluetooth control.

Its dual-rail power system provided stable voltage levels throughout operation, ensuring consistent servo performance and preventing controller resets during extended runs.

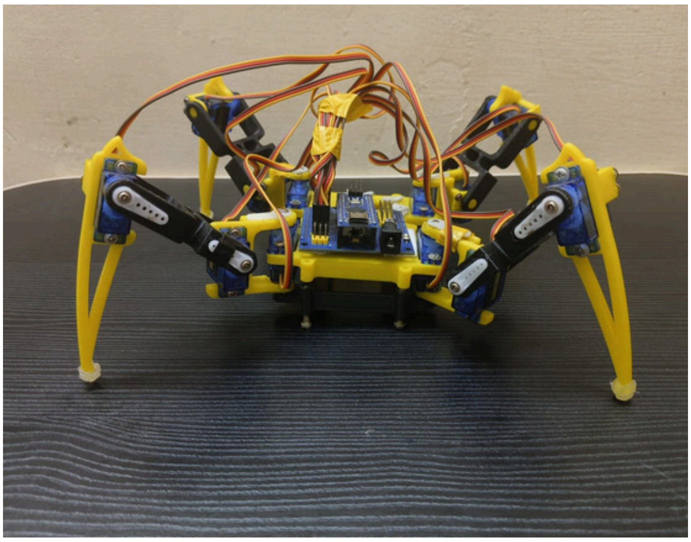

Figure 4.1. Front View of S.P.I.D.E.X

Front View of S.P.I.D.E.X

This image shows the front view of the SPIDEX quadruped robot, highlighting its four-legged structure, servo placements, and compact chassis design. The arrangement ensures mechanical balance and symmetrical weight distribution for stable movement.

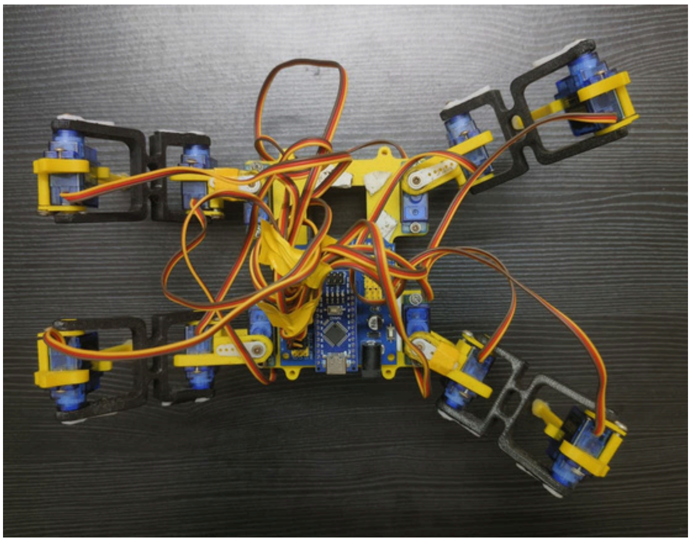

Figure 4.2. Top View and Initial Position of Spider-bot

Top View and Initial Position of Spider-bot

The top view illustrates the initial neutral position of all twelve servos set at 90°, establishing the default stance for calibration. This position ensures that each leg is aligned correctly before executing any motion commands.

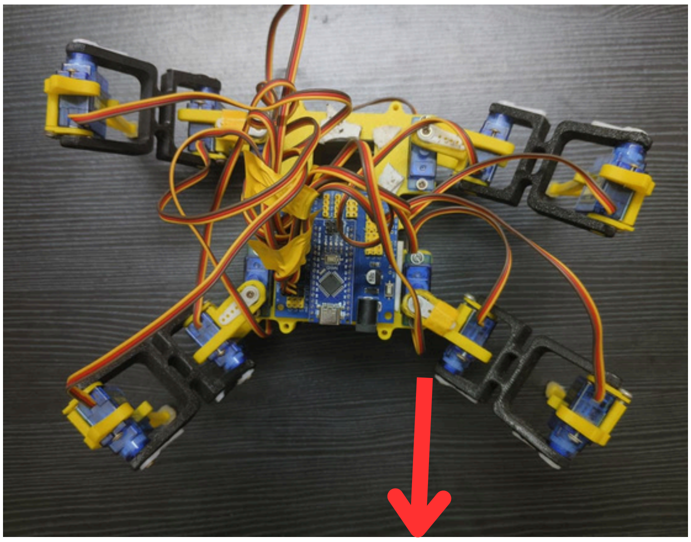

Fig 4.3. Forward Movement

Forward Movement

The figure depicts SPIDEX performing a forward gait sequence where all four legs move cyclically in coordination. The smooth servo transitions validate proper implementation of the inverse kinematics-based walking algorithm.

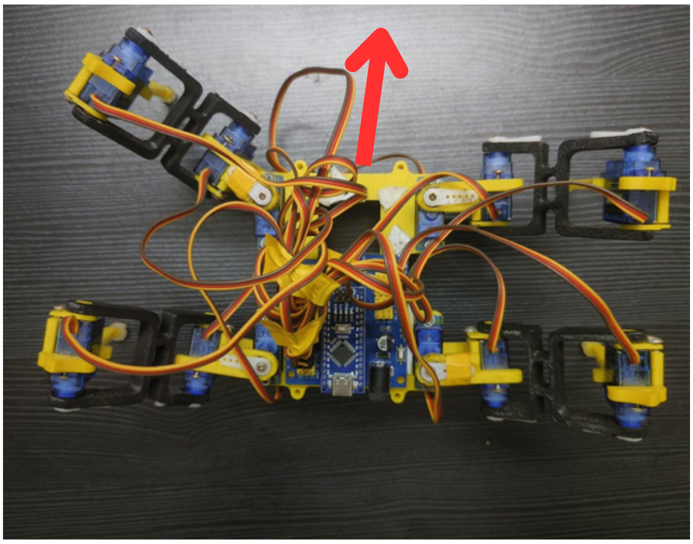

Fig 4.4. Backward Movement

Backward Movement

The figure depicts SPIDEX performing a forward gait sequence where all four legs move cyclically in coordination. The smooth servo transitions validate proper implementation of the inverse kinematics-based walking algorithm.

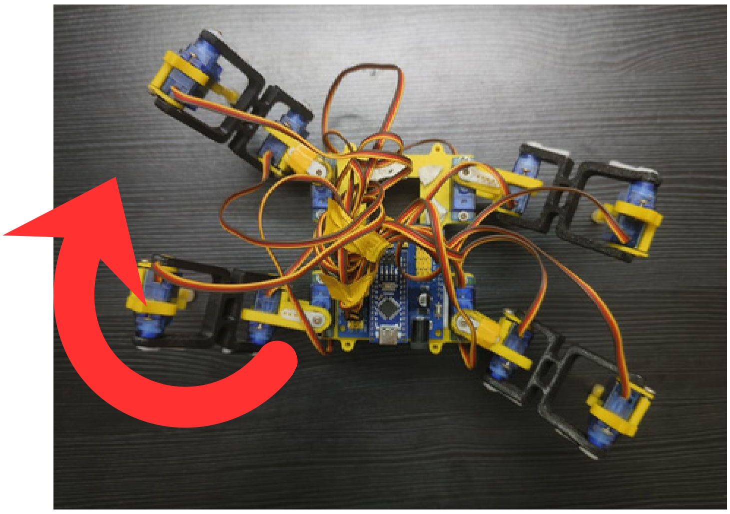

Figure 4.5. Right Turn

Right Turn

The figure shows the robot performing a right-turn maneuver achieved by differential leg movement across the diagonal pairs. The precise servo coordination enables smooth directional change without loss of balance.

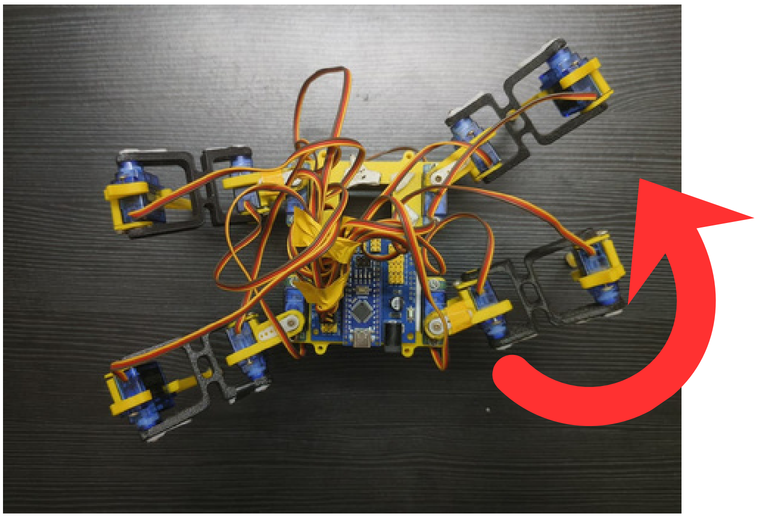

Figure 4.6. Left Turn

Left Turn

This image demonstrates the left-turn movement of SPIDEX, where the diagonal leg pairs alternate to rotate the chassis counterclockwise. The action confirms the effectiveness of the control logic for rotational motion.

4.2. Experimental Evaluation

From the experimental evaluation, it was observed that all motion tests performed by the SPIDEX robot were successful. The robot demonstrated stable and coordinated locomotion in forward, reverse, lateral, and turning directions, confirming the accuracy of the implemented Inverse Kinematics (IK) model and gait algorithm. Each of the twelve servo motors operated smoothly under PWM control without signal lag, ensuring synchronized leg movements throughout the gait cycle. The dual-rail power system, powered by a 7.4V Li-ion battery pack and regulated through an XL4016 buck converter, provided stable voltage to both the Arduino Nano controller and the servo motors. This design prevented brownouts, maintained consistent torque, and ensured uninterrupted operation even during high-current conditions.

The Bluetooth communication link between the HC-05 module and the mobile application was reliable and consistent within an effective range of approximately 10 to 15 meters. The robot responded instantly to directional commands such as forward, backward, left, right, and stop, validating the efficiency of the serial communication protocol and command decoding logic implemented in the Arduino Nano.

4.3. System Efficiency and Overall Performance Analysis

The overall performance of the S.P.I.D.E.X. robot was evaluated based on parameters such as motion stability, response time, power efficiency, and control accuracy. The robot achieved smooth locomotion across all test scenarios, demonstrating efficient power utilization and consistent servo torque output. The dual-rail power supply effectively prevented voltage drops during multi-servo operation, ensuring uninterrupted performance even under load. The Bluetooth communication between the HC-05 module and the mobile application exhibited minimal latency, providing near real-time response to directional commands. The servo motors operated within safe temperature and voltage limits throughout extended testing, confirming the robustness of both the mechanical and electrical systems. These results indicate that SPIDEX successfully meets its design goals in terms of stability, precision, and reliability, making it an effective prototype for further research in robotic locomotion and control.

4.4. Performance Testing and Results

The SPIDEX quadruped robot was tested under various motion conditions to evaluate its stability, accuracy, and overall responsiveness. Each gait pattern — forward, reverse, side walk, turning, and gesture — was executed multiple times using the Bluetooth mobile app interface. The performance results confirmed the proper functioning of the inverse kinematics algorithm, servo synchronization, and power distribution system.

Table 4.1. Experimental Test Results

Test Case	Condition	Observation	Result
Forward Motion	All 4 legs moved cyclically with synchronized servo control.	Smooth gait with minimal jitter and uniform speed.	Successful (✓)
Reverse Motion	Opposite gait sequence of forward motion.	Stable backward motion with proper servo coordination.	Successful (✓)
Side Walk	Alternate diagonal legs operated for lateral movement.	Balanced and steady movement; minor delay due to servo torque at extreme angles.	Successful (✓)
Turning (Left/Right)	Diagonal leg pairs alternated to rotate the body.	Smooth turning with proper balance and no tilt.	Successful (✓)
“Hello” Gesture	One leg raised using three coordinated servos.	Accurate motion with good positional control and stability.	Successful (✓)

CHAPTER 5
CONCLUSION

The project titled “S.P.I.D.E.X – Spider Platform for Integrated Dynamic Engineering X-bot” successfully achieved its aim of designing and developing a functional quadruped robot capable of wireless, app-controlled movement. The system was built using an Arduino Nano microcontroller, HC-05 Bluetooth module, and 12 servo motors, enabling realistic, life-like motion based on inverse kinematic principles. The integration of mechanical, electrical, and programming aspects resulted in a compact, affordable, and efficient robotic platform capable of performing multiple gaits such as walking, turning, and side movement.

A key challenge faced during development was related to power stability. In early stages, powering all servos and logic circuits through a single line caused voltage drops and system resets. This was resolved by implementing a dual-rail power supply using a 7.4V Li-ion battery pack and an XL4016 buck converter, which provided separate regulated power for the control and servo systems. This improvement ensured smooth operation, consistent torque, and prevented controller brownouts, making the robot’s performance reliable and continuous.

The inverse kinematics algorithm was effectively applied to calculate joint angles for each leg, translating user commands into accurate, synchronized servo motion. Each servo was calibrated to a neutral position of 90°, ensuring that the physical movement matched the programmed motion. The robot was successfully tested for various gait patterns, and Bluetooth control worked efficiently within a 10–15 meter range, confirming smooth and responsive operation.

In conclusion, the SPIDEX project fulfilled its objectives by creating a stable, low-cost quadruped robot integrating mechanical design, control logic, and wireless communication. It provided valuable learning in embedded systems, robotics, and motion control, while aligning with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities). SPIDEX serves as a foundation for future developments such as sensor integration, AI-based control, and autonomous movement.

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